3. Write a C program for Playfair algorithm is based on the use of a 5 X 5 matrix of letters constructed using a keyword. Plaintext is encrypted two letters at a time using this matrix.

Aim:

To implement the Playfair cipher using a 5 × 5 matrix constructed from a keyword.

Procedure:

- Start the program.
- Read the keyword and plaintext.
- Construct a 5 × 5 matrix using the keyword, combining I and J.
- Divide the plaintext into pairs of letters.
- Encrypt each pair according to the Playfair rules.
- Display the ciphertext.

Program (C):

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int main() {
    char key[100], text[200], a[26] = "";
    int used[26] = {0}, i, j = 0, k = 0;

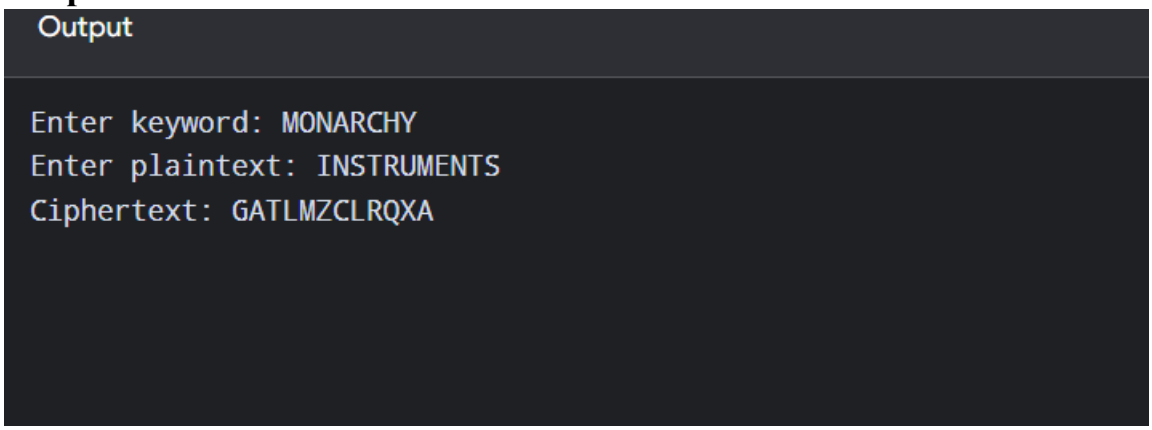
    printf("Enter keyword: ");
    scanf("%99s", key);
    printf("Enter plaintext: ");
    scanf(" %199[^\n]", text);

    for (i = 0; key[i]; i++) {
        char c = toupper(key[i]);
        if (c == 'J') c = 'I';
        if (c >= 'A' && c <= 'Z' && !used[c-'A']) {
            a[j++] = c;
            used[c-'A'] = 1;
        }
    }
```

```
    for (i = 0; i < 26; i++) {
        if (i == ('J'-'A')) continue;
        if (!used[i]) a[j++] = 'A' + i;
    }

    for (i = 0; text[i]; i++) {
        char c = toupper(text[i]);
        if (c >= 'A' && c <= 'Z') {
            if (c == 'J') c = 'I';
            text[k++] = c;
        }
    }
    text[k] = '\0';
    char out[300];
    int n = 0;
    for (i = 0; i < k; i += 2) {
        char x = text[i];
        char y = (i + 1 < k) ? text[i+1] : 'X';
        if (x == y) y = 'X';
        int p = strchr(a, x) - a;
        int q = strchr(a, y) - a;
        int r1 = p / 5, c1 = p % 5;
        int r2 = q / 5, c2 = q % 5;
        if (r1 == r2) {
            out[n++] = a[r1*5 + (c1+1)%5];
            out[n++] = a[r2*5 + (c2+1)%5];
        } else if (c1 == c2) {
            out[n++] = a[((r1+1)%5)*5 + c1];
            out[n++] = a[((r2+1)%5)*5 + c2];
        } else {
            out[n++] = a[r1*5 + c2];
            out[n++] = a[r2*5 + c1];
        }
    }
  out[n] = '\0';
    printf("Ciphertext: %s\n", out);
    return 0;
}
```

Input:
Keyword: MONARCHY
Plaintext: INSTRUMENTS
Output:



Result:

Thus, the Playfair cipher was implemented using a 5 × 5 matrix.